Error in $\begin{cases} \text{outcome} \\ \text{Exposure} \end{cases}$

★ Measurement Error (Attenuation Bias)

- In linear regression, it is assumed that the obs are correctly measured without any error.
- Some variables may not be msbl, then we use proxy variable to measure the effect.

Ex.

$$Y = \beta_0 + \beta_1 A^* + \varepsilon^* \quad (\text{ideal})$$

$$A = A^* + \tau, \quad \underbrace{\tau}_{\text{random disturbance}} \sim (0, \sigma_\tau^2)$$

Rewrite
$$\begin{aligned} Y &= \beta_0 + \beta_1 (A - \tau) + \varepsilon^* \\ &= \beta_0 + \beta_1 A + (\varepsilon^* - \beta_1 \tau) \\ &= \beta_0 + \beta_1 A + \varepsilon \end{aligned}$$

Hence
$$\begin{aligned} \mathbb{E}(A\varepsilon) &= \mathbb{E}[(A^* + \tau)(\varepsilon^* - \beta_1 \tau)] \\ &= -\beta_1 \mathbb{E}(\tau^2) = -\beta_1 \sigma_\tau^2 \neq 0 \end{aligned}$$

$$\check{\beta}_1 = \frac{Cov(Y, A)}{V(A)} = \frac{Cov(\beta_0 + \beta_1 A^* + \varepsilon^*, A^* + \tau)}{V(A)}$$

$$= \beta_1 \left[ \underbrace{\frac{V(A^*)}{V(A^*) + V(\tau)}}_{<1} \right] < \beta_1$$

★ Simultaneous Bias

- Simultaneity arise when one or more of the preditors is determined by response variable.
- In simple terms

$$A \rightleftarrows Y \quad \text{Reverse Causation}$$

A cause Y and Y cause A.

Ex.

$$\begin{aligned} Y &= \alpha X + u \\ X &= \beta Y + \nu \end{aligned}$$ by solving equations $\Rightarrow$ Reduced form $Y = \frac{\alpha\nu + u}{1 - \alpha\beta}, \quad X = \frac{\alpha u + \nu}{1 - \alpha\beta}$

$$Cov(X,u) = Cov\left(\frac{\alpha u + v}{1-\alpha\beta}, u\right) = \frac{\alpha}{1-\alpha\beta} V(u) \neq 0$$

It suffer endogeneity issue.

Ex. ( Omitted Variable)

$Y$: family income $\quad \ln(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

$X_1$: Husband Education $\quad \ln(Y) = \beta_0 + \beta_1 X_1 + \tau$ (Omit $X_2$)

$X_2$: Wife "

| | Model 1 | Model 2. |
|---|---|---|
| Const | 10.264 | 10.539 |
| $X_1$ | 0.0439 | 0.0613 |
| $X_2$ | 0.0390 | |

The effect of $X_2$ has been incorrectly attributed to the $X_1$ - leading to an overstatement of the latter's importance.

Ex. (Measurement Error)

$Y$: Colleage GPA

$C_1$: high school GPA

$C_2$: SAT

$$\underbrace{A}_{\text{report by student.}} = \underbrace{A^*}_{\text{actual family annual income}} + \tau$$

$$Y = \beta_0 + \beta_A A^* + \beta_1 C_1 + \beta_2 C_2 + \varepsilon^*$$

Using $A$ in place of $A^*$ will bias the OLS estor of $\beta_A$ toward zero.

Remark.

The effect of biasing the coef toward to zero is called attenuation.

Ex. (Simultaneous Bias)

$Y$: murders per capita

$X$: No. of police officers per capita

$Z$: income per capita

- Cities often want to determine how much additional law enforcement will decrease their murders rates.

$$Y = \alpha_1 X + \alpha_2 Z + u$$

- Can we think of police force size as being exogenously determined?

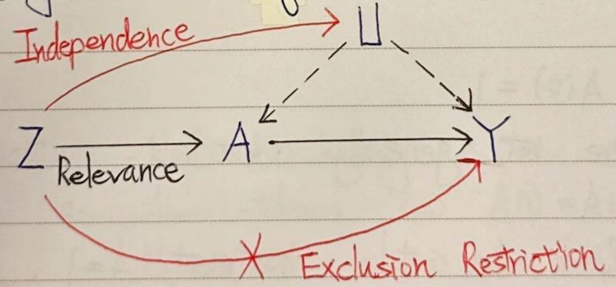
Probably not.

- A city's spending on law enforcement is at least partly determined by its expected murder rates.

$$X = \beta_1 Y + v$$

☆ What is remedy of Endogeneity?

⇒ Introducing one or more instrument variables via Two-Stage Least Square (2SLS)




Def. A variable $Z$ is instrument because it meet

1. Relevance, $Cov(Z, A) \neq 0$ or $Z \not\perp\!\!\!\perp A$
   ($Z$ is associated w/ $A$)
2. Exclusion Restriction, $Y(a, z) = Y(a, z^*) = Y(a)$ $\forall z$ and $z^*$ $\forall a$
   ($Z$ does not affect $Y$ except through its potential effect on $A$)
3. Independence, $Z \perp\!\!\!\perp U$
   ($Z$ is indep of $U$ that affect both $A$ and $Y$)

The assumption 1 ~ 3 are insufficient for identification

★ 4th IV assumption

a. Effect Homogeneity, $\underbrace{Y(1)}_{A=1} - \underbrace{Y(0)}_{A=0} \perp\!\!\!\perp Z$

(Constant effect of A on Y across value of covariates)

b. Monotonicity, $\underbrace{A(1)}_{Z=1} \geq \underbrace{A(0)}_{Z=0}$ for all individuals (No Defiers)

(Z shift A (in same) direction for all individuals)

★ If we know the value of $A(1)$ and $A(0)$ for each individuals we could classify all individuals into 4 disjoint subpopulations

principal strata

1. Always-takes,

$$A(1) = A(0) = 1$$

2. Never-takes

$$A(1) = A(0) = 0$$

3. Compliers

$$A(1) = 1, \; A(0) = 0$$

4. Defiers

$$A(1) = 0, \; A(0) = 1$$

these subpopulation are not generally identified.

- Scenario 1.

  Assign to $Z=1$ and take treatment $A=1$, $A(1)=1$

  Complier or Always-taker?

- Scenario 2.

  Assign to $Z=1$ and take treatment $A=0$, $A(1)=0$

  Defier or Never-taker?

★ Identification by Effect Homogeneity

$\Updownarrow$

Constant Effect

$$Y(1) - Y(0) = \beta_0 \Rightarrow Y(1) = Y(0) + \beta_0$$

Observed $Y = \begin{cases} Y(1) & \text{if } A=1 \\ Y(0) & A=0 \end{cases}$

$$= Y(0) + \{Y(1) - Y(0)\} A$$
$$= Y(0) + \beta_0 A$$

When $Z$ is instrument, under Exclusion Restriction

We have $Y(a) \perp\!\!\!\perp Z$

$\Rightarrow E(Y(0) \mid Z=1) = E(Y(0) \mid Z=0)$

It implies

$$E(Y - \beta_0 A \mid Z=1) = E(Y - \beta_0 A \mid Z=0)$$

$$\Rightarrow E(Y \mid Z=1) - \beta_0 E(A \mid Z=1)$$
$$= E(Y \mid Z=0) - \beta_0 E(A \mid Z=0)$$

$$\Rightarrow \beta_0 \{E(A \mid Z=1) - E(A \mid Z=0)\} = E(Y \mid Z=1) - E(Y \mid Z=0)$$

$$\Rightarrow \beta_0 = \frac{E(Y \mid Z=1) - E(Y \mid Z=0)}{E(A \mid Z=1) - E(A \mid Z=0)}$$

which is the usual IV estimand for dichotomous Instrument.

★ Identification by Monotonicity

Let $A$ as Always-takers $\Leftrightarrow A(1) = A(0) = 1$

$N$ Never-takers $\Leftrightarrow A(1) = A(0) = 0$

$C$ Compliers $\Leftrightarrow A(1) = 1, A(0) = 0$

$D$ Defiers $\Leftrightarrow A(1) = 0, A(0) = 1$

We can decompose causal effect on $Y$ into 4 terms

$$E[Y_{(Z=1)} - Y_{(Z=0)}] = E[Y_{(Z=1)} - Y_{(Z=0)} \mid A] P(A)$$
$$+ E[\quad " \quad \mid N] P(N)$$
$$+ E[\quad " \quad \mid C] P(C)$$
$$+ E[\quad " \quad \mid D] P(D)$$

It is weighted average of 4 latent subgroup effects.

i) Given the Exclusion Restriction

$$E[Y(Z=1) - Y(Z=0)] = E[Y(Z=1, A(1)) - Y(Z=0, A(0))]$$
$$= E[Y(A(1)) - Y(A(0))]$$

Thus.

$$E[Y(Z=1) - Y(Z=0) \mid A) = 0$$
$$E[Y(Z=1) - Y(Z=0) \mid N) = 0$$

ii) Given Monotonicity to rule out Defiers

$$E[Y(Z=1) - Y(Z=0)] = E[Y(1) - Y(0) \mid C) P(C)$$

$A(1)=1, A(0)=0$

iii) Identification of P(C)

Consider causal effect Z on A

$$\begin{aligned} E[A(1) - A(0)] &= E[A(1) - A(0) \mid A] P(A) \\ &+ E[\quad \text{''} \quad \mid N] P(N) \\ &+ E[\quad \text{''} \quad \mid C] P(C) \\ &+ E[\quad \text{''} \quad \mid D] P(D) \\ &= P(C) \end{aligned}$$

$$\left( \begin{aligned} &= 1 - P(A=0 \mid Z=1) - P(A=1 \mid Z=0) \\ &= 1 - \{1 - P(A=1 \mid Z=1)\} - P(A=1 \mid Z=0) \end{aligned} \right)$$

Under independence assumption

$$E[A(1) - A(0)] = P(A(1) - A(0) = 1)$$
$$\triangleq P(A=1 \mid Z=1) - P(A=1 \mid Z=0)$$
$$E[Y(Z=1) - Y(Z=0)] \triangleq E(Y \mid Z=1) - E(Y \mid Z=0)$$

iv) by i) ~ iii)

$$E[Y(1) - Y(0) \mid C] = \frac{E(Y \mid Z=1) - E(Y \mid Z=0)}{P(A=1 \mid Z=1) - P(A=1 \mid Z=0)}$$

We idenfy the effect as the ratio of the diff in means of outcome over diff in means of treatment received.

(Remark. usual IV estimand equals the ave causal effect in Complier under monotonicity.
$$E[Y(1) - Y(0) \mid C]$$ )

Village Level

(CRT) 集群隨机試驗

Ex. (Somer–Zeger Vitamin)

- Community-based, cluster-randomized trial in Indonesia.
- Evaluate the impact of high-dose vitamin A supplementation on infant survial. → Outcome
- No infant in control villiage received supplement.

Data w/ $n = 23682$

| Z | A | Y | No. units |
|---|---|---|---|
| 0 | 0 | 0 | 74 |
| 0 | 0 | 1 | 11514 |
| 1 | 0 | 0 | 34 |
| 1 | 0 | 1 | 2385 |
| 1 | 1 | 0 | 12 |
| 1 | 1 | 1 | 9663 |

$Y = 1$ if the infant survived at follow-up period
$A = 1$ 〃 received the supplement.
$Z = 1$ 〃 villiage was randomized to receive treatment.

- Analysis

$$P(Y=1 \mid Z=1) = \frac{2385 + 9663}{34 + 2385 + 12 + 9663} = \frac{12048}{12094}$$

$$P(Y=1 \mid Z=0) = \frac{11514}{74 + 11514} = \frac{11514}{11588}$$

$$P(A=1 \mid Z=1) = \frac{12 + 9663}{12094} = \frac{9675}{12094}$$

$$P(A=1 \mid Z=0) = \frac{0}{11588} = 0$$

then

$$\frac{P(Y=1 \mid Z=1) - P(Y=1 \mid Z=0)}{P(A=1 \mid Z=1) - P(A=1 \mid Z=0)} = \frac{\frac{12048}{12094} - \frac{11514}{11588}}{\frac{9675}{12094} - 0} = \frac{0.258\%}{80\%}$$

the remaining 20% are non-compliers

$$\approx 0.323\%$$

$\Rightarrow$ Reduction in infant death of 0.323% ← every 1000 complier infants about +3.23 infants are expected survive due to intervention.

## ≷ IV in LM.

- In practice, a valid IV may consist of multiple variables or be conti or multi-level.

Consider
$$Y = \beta_0 + \beta_a A + \beta_u U + \varepsilon$$
$$Y = \beta_0 + \beta_a A + \varepsilon^* \quad \text{(Fitted)}$$
$$\varepsilon^* = \beta_u U + \varepsilon$$
$$Cov(Y, Z) = \beta_a Cov(A, Z) + Cov(\varepsilon^*, Z)$$

then
$$\frac{Cov(Y,Z)}{Cov(A,Z)} = \beta_a + \frac{Cov(\varepsilon^*, Z)}{Cov(A,Z)}$$
$(Z \perp\!\!\!\perp \varepsilon^*$ by Exclusion Restriction$)$
$$= \beta_a$$

(Recall that in Simple linear Reg
$$Y = \beta_0 + \beta_1 X + \varepsilon$$
$$\beta_1 = \frac{Cov(Y,X)}{V(X)}$$
)

$$Y = \beta_0 + \beta_a[\pi_0 + \pi_Z Z + \nu] + \varepsilon^*$$
$$= \theta_0 + \theta_Z Z + u \quad \text{(reduced)}$$

$$\beta_a = \frac{\frac{Cov(Y,Z)}{V(Z)}}{\frac{Cov(A,Z)}{V(Z)}} = \frac{\theta_Z}{\pi_Z}$$

$$A = \pi_0 + \pi_Z Z + \nu$$

☆ Two Stage Least Square (2SLS)

- A computational method for calculating the causal effect in two-stage proc.

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- The underlying true model are assumed to be linear

$$Y = \beta_0 + \beta_a A + \beta_u U + \varepsilon$$
$$A = \pi_0 + \pi_z Z + \pi_u U + \nu$$

★ Two-Stage Proc.

Stage 1: $A \sim Z$

$$A = \pi_0^A + \pi_z^A Z + \nu_1 \Rightarrow \text{get } \hat{A}$$

Stage 2: $Y \sim \hat{A}$

$$Y = \pi_0^Y + \pi_a^Y \hat{A} + \nu_2 \Rightarrow \text{get } \hat{\pi}_a^Y$$

★ Why 2SLS work?

$$E(Y|Z) = E(\beta_0 + \beta_a A + \beta_u U + \varepsilon | Z)$$
$$= \beta_0 + \beta_a E(A|Z) + \beta_u E(U|Z) + E(\varepsilon|Z)$$
$$= \beta_0 + \beta_a E(A|Z)$$

★ General Model

$A_1, \dots, A_k$ Endogenous regressors → potentially correlated w/ $\varepsilon$

$X_1, \dots, X_r$ Included exogenous variables → uncorrelated w/ $\varepsilon$

$Z_1, \dots, Z_m$ Instrumental variables → Excluded exogenous variables

Def. A parameter is said to be identified if different values of parameter will produce different distⁿ of the data.

In IV regression, the coefs are identified depends on the relation between $\begin{cases} \text{No. of instruments } (m) \\ \text{No. of endogenous regressor } (k) \end{cases}$

$$Y = \beta_0 + \beta_1 A_1 + \cdots + \beta_k A_k + \theta_1 X_1 + \cdots + \theta_r X_r + \varepsilon$$

The coef $\beta_1, \dots, \beta_k$ are said to be

i) exact identified if $m = k$ (just enough to est'é)

ii) overidentified " $>$

iii) underidentified " $<$ (Too few to est'é)

1. The $k$ 1st stage model, $j=1,\dots,k$

$$A_j = \alpha_0^j + \alpha_1^j X_1 + \cdots + \alpha_r^j X_r + \gamma_1^j Z_1 + \cdots + \gamma_m^j Z_m + \nu^j$$

obtain $\hat{A}_1, \dots, \hat{A}_k$

2. In 2nd Stage

$$Y = \beta_0 + \beta_1 \hat{A}_1 + \cdots + \beta_k \hat{A}_k + \theta_1 X_1 + \cdots + \theta_r X_r + \varepsilon^*$$

☆ Projection Viewpoint.

Let $\underset{\sim}{y}$ : $n\times 1$ Outcome vector

$\mathbb{A}$ : $n\times(k+1)$ Endogenous regressors

$\mathbb{X}$ : $n\times r$ Exogenous regressors

$$\underset{\sim}{y} = \mathbb{A}\underset{\sim}{\beta} + \mathbb{X}\underset{\sim}{\theta} + \underset{\sim}{\varepsilon}$$

$$= \mathbb{D}\underset{\sim}{\delta} + \underset{\sim}{\varepsilon}$$

$\mathbb{Z}$ : $n\times m$ Instrumental variables

$\mathbb{W} = [\mathbb{X}\ \mathbb{Z}]$ : $n\times(r+m)$ full set of instruments

i) $\mathbb{A} \sim \mathbb{W} \Rightarrow$ get $\hat{\mathbb{A}}$

$$\hat{\mathbb{A}} = \mathbb{W}(\mathbb{W}^T\mathbb{W})^{-1}\mathbb{W}^T\mathbb{A} \quad \left(\begin{array}{l}\text{Project } A \text{ onto space} \\ \text{spanned by } \mathbb{W}\end{array}\right)$$

$$= \mathbb{P}\mathbb{A}$$

ii) $\underset{\sim}{y} \sim \hat{\mathbb{A}}$ and $\mathbb{X}$

$$\hat{\underset{\sim}{\delta}} = \left([\mathbb{A}\ \mathbb{X}]^T\mathbb{P}[\mathbb{A}\ \mathbb{X}]\right)^{-1}[\mathbb{A}\ \mathbb{X}]^T\mathbb{P}\underset{\sim}{y}$$

$$= \left(\mathbb{D}^T\mathbb{P}\mathbb{D}\right)^{-1}\mathbb{D}^T\mathbb{P}\underset{\sim}{\varepsilon}$$

$$= \underset{\sim}{\delta} + (\mathbb{D}^T\mathbb{P}\mathbb{D})^{-1}\mathbb{D}^T\mathbb{P}\underset{\sim}{\varepsilon}$$

→ unbiased if IVs are valid

Let $\mathbb{H} = \mathbb{P}\mathbb{D}$

$$\hat{\underset{\sim}{\delta}} = (\mathbb{H}^T\mathbb{D})^{-1}\mathbb{H}^T\underset{\sim}{y}$$

$$= \underset{\sim}{\delta} + (\mathbb{H}^T\mathbb{D})^{-1}\mathbb{H}^T\underset{\sim}{\varepsilon}$$